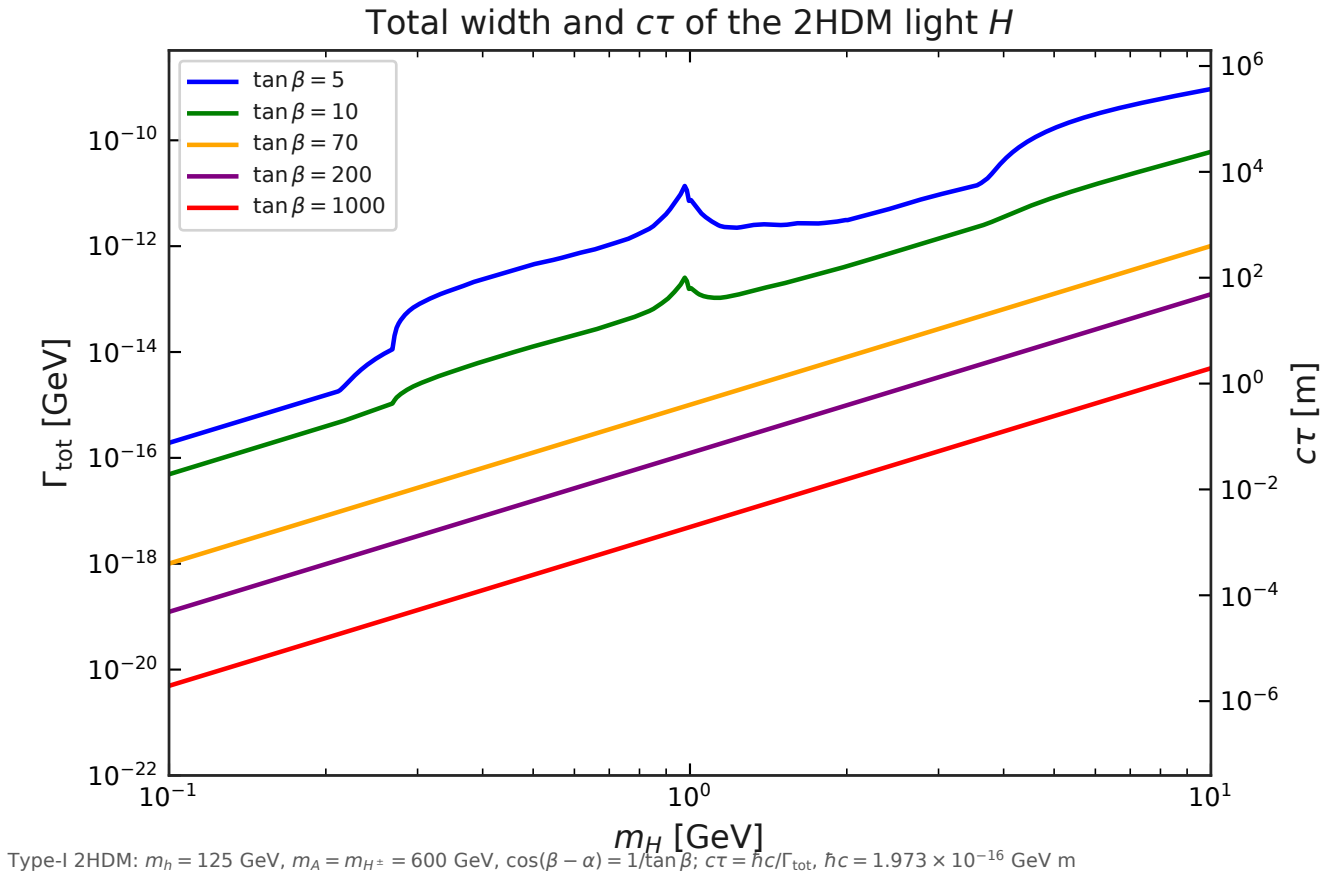
Total width and $c\tau$ of the 2HDM light H
$\tan\beta = 5$
$\tan\beta = 10$
$\tan\beta = 70$
$\tan\beta = 200$
$\tan\beta = 1000$
Γ_{tot} [GeV]
$c\tau$ [m]
m_H [GeV]
10^{-10}
10^{-12}
10^{-14}
10^{-16}
10^{-18}
10^{-20}
10^{-22}
10^{6}
10^{4}
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-1}
10^{0}
10^{1}
Type-I 2HDM: $m_h = 125$ GeV, $m_A = m_{H^\pm} = 600$ GeV, $\cos(\beta - \alpha) = 1/\tan\beta$; $c\tau = \hbar c/\Gamma_{\text{tot}}$, $\hbar c = 1.973 \times 10^{-16}$ GeV m